DAVID IZUCHUKWU GILBERT

Chemical Engineering Student | Self-Taught Software Developer
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PROFILE

Chemical Engineering undergraduate at the University of Lagos and self-taught software developer. Builds and maintains student software, practical web interfaces and engineering prototypes, independently and with small teams. Official CGPA: 4.94/5.00 through 100 level; updated 200 level results are awaiting publication.

EDUCATION

Bachelor of Engineering, Chemical Engineering

University of Lagos | 2024-2029 expected | 200 Level

- Coursework includes thermodynamics, material and energy balances, fluid mechanics, heat transfer, engineering materials, mathematics and laboratory work.

Senior Secondary Education

Elite Crop College | 2020-2023

- Overall Best Senior Secondary School Student and Head Boy, 2023.
- SAT: 1360/1600; JAMB UTME: 302/400 (2024). WAEC: three A grades; NECO: five A grades across nine subjects each (2023).

EXPERIENCE AND TRAINING

SIWES Trainee, Process Department (SM6)

Nigerite | September 2026-present | On-site from 5 October 2026

- Under supervision, collect and document production records from pre-autoclaving and the wet machine; check material quantities and report daily vacuum and injector readings to the supervisor.
- Performed premixer/mixer testing, viscosity and moisture measurements with guidance; observed production and curing operations and assisted with pipe cleaning and preparation work. Placement ongoing.

Backend Intern

HNG Internship Cohort 13 | October-December 2025

- Completed Stages 0-3 through time-bound backend tasks involving APIs, testing, documentation and version-controlled delivery.

AI/ML Track Participant

ECX UNILAG | January 2025-present

- Undertook applied machine-learning training and project work, including credit-card fraud detection.

AI/ML Fellow

3 Million Technical Talent (3MTT), Cohort 4 | Ongoing

- Participating in the AI/ML pathway.

GCI World 2026 September Participant

Matsuo-Iwasawa Laboratory, University of Tokyo | September 2026-present

- Accepted into the ongoing data science and machine-learning programme; coursework in progress.

Operator-Training Simulation Practice

National RTSim OTS Workshop and RTsim.Career | 2026

- Practising process start-up, control-loop operation and troubleshooting in buffer-tank, heat-exchanger, amine-treatment and distillation simulations.

SOFTWARE AND MACHINE LEARNING PROJECTS

Swift | Owner and Full-Stack Developer

Student-learning platform | 2023-present | Live site | Repository

- Independently rebuilt and now maintain the platform after its initial collaborative phase; currently used by several students.
- Built UNILAG Chemical Engineering course-material, test, authentication, account-recovery and CGPA workflows, extending coverage through the CCMAS 300 level curriculum.
- Built AI-assisted question-generation workflows using FastAPI/Python, Supabase-backed material extraction and LLM integrations.

Credit-Card Fraud Detection

ECX UNILAG AI/ML Track | Repository

- Built and evaluated models on approximately 568,000 transactions, recording 99.97% recall on a separate test set.
- Applied feature engineering, SMOTE, Random Forest and XGBoost with Python, pandas and scikit-learn.

Multilingual Nigerian News Classification and Headline Generation

DSN Bootcamp Hackathon 2026 | LLM/Agent Track

- Developed and evaluated approaches for Hausa, Igbo, Yoruba and Nigerian Pidgin.
- Experimented with mT5 and LoRA fine-tuning, TF-IDF/LinearSVC classification and extractive headline baselines; prepared competition notebooks and submissions.

Vouch | Frontend Engineer

Squad Hackathon 3.0 | 63rd of 1,600 entries | Live site | Repository

- Built trader-facing interfaces for a portable business trust-profile product, including dashboard, score visualisation, tier progression and live Server-Sent Events updates.
- Continue frontend, product and documentation work with the team; the scoring model and backend were developed separately.

CrxRay | Hackathon Project

Tech Literacy Network Hackathon 2026 | Repository

- Built a React browser-extension risk scanner presenting permission and privacy signals as A-F assessments with plain-language explanations.
- Prepared the working demo, repository documentation and hackathon submission.

Resonance | Frontend Engineer

DSN x BCT LLM Agent Challenge | Four-person team | Live site

- Developed the interface for a two-agent product-intelligence system; integrated Firebase Authentication and Firestore to preserve prediction history.

NexStore UI | Frontend Project

Live site | Repository

- Built a storefront frontend with responsive layouts and reusable components. Further pages and functional improvements remain in development.

Urban Brand Creatives | Frontend Design Project

Live site

- Designed a business website interface originally intended for a family member. The business did not adopt it; retained as a frontend design project.

Pepper Table and Commonwell Church | Website Concepts

Pepper Table | Commonwell Church

- Built and deployed a restaurant website concept with sample menu search and filtering, and an imagined church website concept. Neither represents an operating client engagement.

Neuronode | Solo Developer

BuildQuik Challenge | Live site

- Built a DSA tutor with guided questioning, stuck-point detection and learning/testing modes for array problems using React, Node.js and Express.

ENGINEERING PROJECTS AND COMPETITIONS

Cement Production Conversion Gap Analysis | Team Leader

Dangote Cement Plc University Engineering Challenge 2026 | Team Kilvora | Top 20 of 300+ submissions

- Led ANSYS structural and thermal analysis of a mill-shell and liner model, with supporting calculations, scalability analysis and technical research.
- Contributed to team analysis estimating a 27.6% conversion gap and modelling a pathway to 85.0% conversion; these are model estimates, not plant results.
- Prepared technical and business-case materials with a two-person team.

TLUD Gasifier Cookstove and PAYG Adoption Model | Team Leader

The Industry Discourse 5.0 Energy Startup Challenge | Team Axel | Top 20 semifinalist

- Directed a four-person team developing a design-stage Top-Lit Updraft gasifier cookstove concept paired with a pay-as-you-go adoption model; reviewed fuel, market and financial assumptions.

Klova Passive Biofiltration Cartridge | Team Leader

PIDEC 1.0 | Team CHG | 4th among 10 departmental teams

- Originated the concept and led a six-person team through design revisions, laboratory preparation, prototype construction and testing.
- Built a gravity-fed test arrangement and cartridge using mesh, charcoal/biochar, granite and soil, incorporating Bacillus and Pseudomonas under laboratory guidance.
- Created two Blender process visualisations. Stage 1 score: 90.5/100, the highest Chemical Engineering score.

GraphSense Accessible Chart-to-Audio Prototype | MATLAB Developer

AFRETEC-UNILAG MathWorks category | Team Sync | 1st place

- Developed the MATLAB implementation and demo video for a four-person team project based on Mumeen's original concept.
- Built an interface that maps chart data to changing audio pitch, strongest for line graphs; conducted informal blindfold trials with sighted participants.
- A separate web beta is in progress at graphsense-web.vercel.app.

TECHNICAL SKILLS

Software: React, Next.js, JavaScript, TypeScript, HTML, CSS, Tailwind CSS, Vite, Node.js, Express, FastAPI, Python, REST APIs, PostgreSQL, Supabase, Firebase, Firestore, Prisma ORM, Git and GitHub.

Machine learning: pandas, NumPy, scikit-learn, XGBoost, Random Forest, SMOTE, Hugging Face Transformers, LoRA, mT5, ROUGE and BERTScore.

Engineering and design: ANSYS, MATLAB, JMP, RTsim.Career, AutoCAD, Fusion 360 and Blender; mass and energy balances, heat transfer and process-control fundamentals.

CERTIFICATIONS, MEMBERSHIPS AND RECOGNITION

Certificates: AI/ML Essentials, ECX 6.0 Engineering Career Expo; Generative AI Foundation, Medium and Master, UNICEF / Nigeria Learning Passport; Tech Literacy Network Hackathon 2026 participation.

Memberships: Student member, American Institute of Chemical Engineers (AIChE) and Nigerian Society of Chemical Engineers (NSChE).

Recognition: GraphSense 1st place, AFRETEC-UNILAG MathWorks category; DCP Top 20 of 300+; PIDEC 4th of 10; TID 5.0 Top 20 semifinalist; Vouch 63rd of 1,600; Overall Best Senior Secondary School Student, 2023.

Leadership: Head Boy, Elite Crop College, 2023; team leader for Team Kilvora, Team Axel and Team CHG, 2026.